

CS232/CS332 Introduction to Cloud Computing Technology

Module 06: Compute (Part 1)

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Lesson Learning Outcomes

After this module ends, students must be able to:

- Describe various AWS compute services, including Amazon EC2, AWS Lambda, AWS Elastic Beanstalk, and container services.
- Demonstrate the ability to launch, configure, and manage Amazon EC2 instances using the AWS Management Console.
- Identify cost optimization strategies for Amazon EC2, including choosing the appropriate pricing model (On-Demand, Reserved, Spot, etc.) and right-sizing instances.
- Explain the benefits of AWS Lambda, create and deploy Lambda functions, and integrate them with AWS services.
- Demonstrate how to deploy and manage web applications using AWS Elastic Beanstalk, understanding its automation capabilities for scaling and deployment.

จากแบบทดสอบก่อนเรียน M6

- ยังสับสนเรื่องการเลือกบริการ Compute ให้เหมาะกับงานแต่ละแบบ โดยเฉพาะความแตกต่างระหว่าง EC2, Container และ Lambda ว่าควรใช้ในกรณีไหนค่ะ
- ใน lab แรกของ module ตอนสร้าง launch คือรันได้แต่ไม่ผ่านในส่วนของ key checkค่ะ
- ลำดับขั้นตอนบางส่วน เช่น การต้อง Stop instance ก่อนปรับขนาด และการเปลี่ยนแปลงของ Public IP หลังจาก Stop และ Start ใหม่ ซึ่งต้องใช้เวลาในการทำความเข้าใจมากขึ้น
- ขั้นตอน monitor instance เพราะเป็นหน้าจอ command แล้วมีอะไรขึ้นมาไม่รู้เยอะแยะเต็มไปหมด
- การตั้งค่า Security Group เพราะต้องเลือกพอร์ตและ source ให้ถูก ถ้าตั้งค่าผิดก็จะเข้าเว็บเซิร์ฟเวอร์ไม่ได้ ทำให้ตอนแรกไม่แน่ใจว่าควรเปิดพอร์ตอะไรและเปิดให้ใครเข้าได้บ้าง
- การตั้งค่า Security Groups เพื่อเข้าถึง EC2 เพราะยังสับสนนิดๆเรื่องการเปิด Port และการเชื่อมโยงสิทธิ์การเข้าถึงระหว่าง Firewall กับตัวเครื่อง
- ยังไม่เข้าใจ instance typeว่าการแบ่งเป็น categories แล้วส่งผลหรือใช้งานต่างกันยังไง
- อยากให้อาจารย์อธิบายจุดเด่นของแต่ละ service อีกรอบหนึ่งเป็นการเน้นย้ำความเข้าใจ



Module 6: Compute

AWS Academy Cloud Foundations

Module overview

Topics

- Compute services overview
- Amazon EC2
- Amazon EC2 cost optimization
- Container services
- Introduction to AWS Lambda
- Introduction to AWS Elastic Beanstalk

Activities

- Amazon EC2 versus Managed Service
- Hands-on with AWS Lambda
- Hands-on with AWS Elastic Beanstalk

Demo

- Recorded demonstration of Amazon EC2

Lab

- Introduction to Amazon EC2



**Knowledge
check**

Section 1: Compute services overview

Module 6: Compute

AWS compute services

Amazon Web Services (AWS) offers many compute services. This module will discuss the highlighted services.



Amazon EC2



Amazon EC2
Auto Scaling



Amazon Elastic
Container Registry
(Amazon ECR)



Amazon Elastic
Container Service
(Amazon ECS)



VMware Cloud
on AWS



AWS Elastic
Beanstalk



AWS Lambda



Amazon Elastic
Kubernetes Service
(Amazon EKS)



Amazon Lightsail



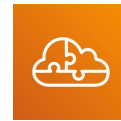
AWS Batch



AWS Fargate



AWS Outposts



AWS Serverless
Application Repository

Categorizing compute services

Services	Key Concepts	Characteristics	Ease of Use
Amazon EC2	- Infrastructure as a service (IaaS) - Instance-based Virtual machines	Provision virtual machines that you can manage as you choose	A familiar concept to many IT professionals.
AWS Lambda	Serverless computing - Function-based - Low-cost	- Write and deploy code that runs on a schedule or that can be triggered by events - Use when possible (architect for the cloud)	A relatively new concept for many IT staff members, but easy to use after you learn how.
- Amazon ECS - Amazon EKS - AWS Fargate - Amazon ECR	Container-based computing - Instance-based	Spin up and run jobs more quickly	AWS Fargate reduces administrative overhead, but you can use options that give you more control.
AWS Elastic Beanstalk	- Platform as a service (PaaS) - For web applications	- Focus on your code (building your application) - Can easily tie into other services—databases, Domain Name System (DNS), etc.	Fast and easy to get started.

Choosing the optimal compute service

- The optimal compute service or services that you use will depend on your use case
- Some aspects to consider —
 - What is your application design?
 - What are your usage patterns?
 - Which configuration settings will you want to manage?
- Selecting the wrong compute solution for an architecture can lead to lower performance efficiency
 - A good starting place—Understand the available compute options

Activity 1:

"Mentimeter Compute Service Quizzes"

(15 minutes)

Section 2: Amazon EC2

Module 6: Compute



Amazon Elastic Compute Cloud (Amazon EC2)

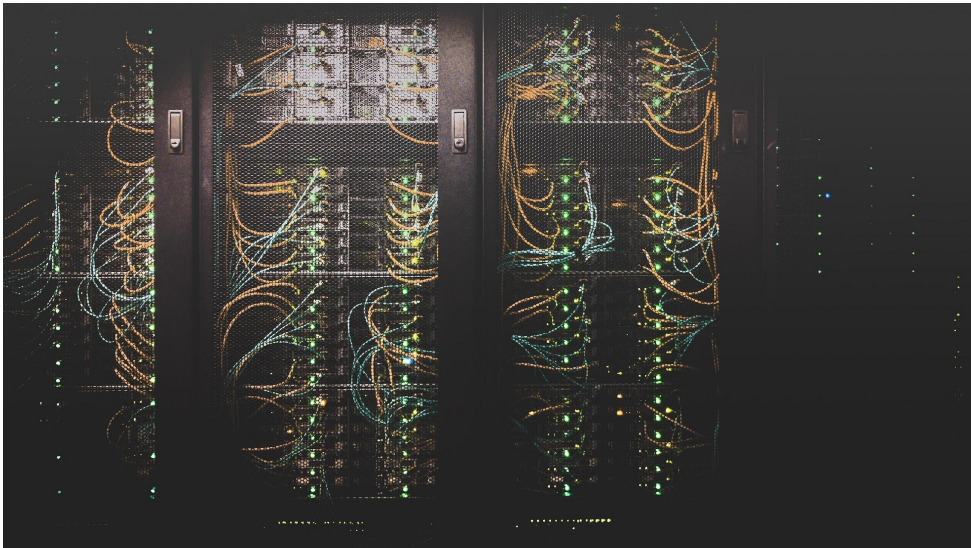
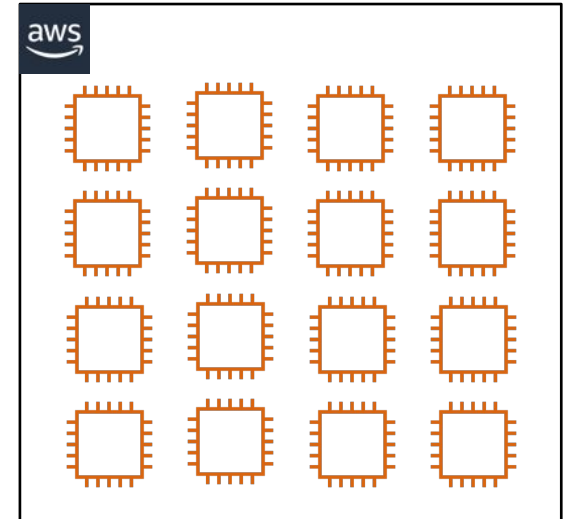


Photo by Taylor
Vick on Unsplash

**On-premises
servers**

Example uses of Amazon EC2 instances

- ✓ Application server
- ✓ Web server
- ✓ Database server
- ✓ Game server
- ✓ Mail server
- ✓ Media server
- ✓ Catalog server
- ✓ File server
- ✓ Computing server
- ✓ Proxy server

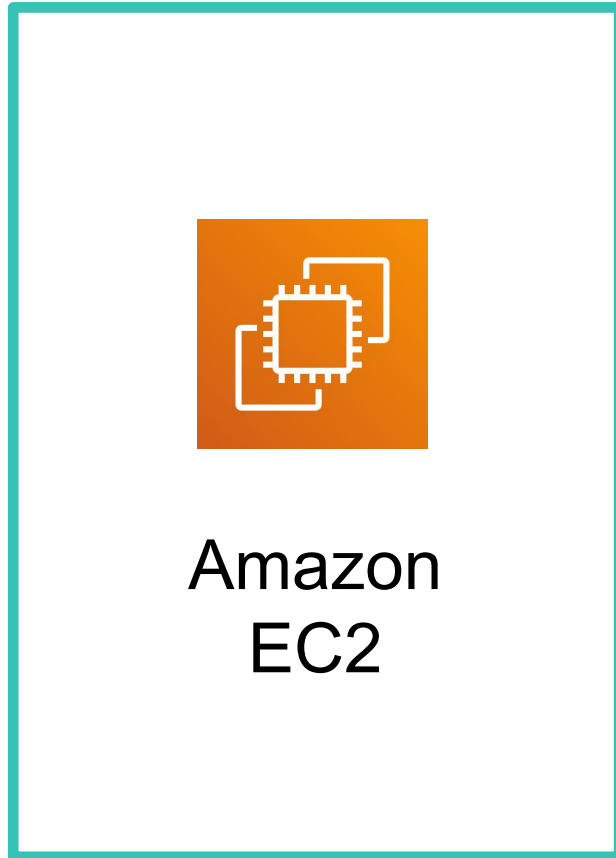


Amazon EC2 instances



Photo by panumas nikhomkhai
from Pexels

Amazon EC2 overview

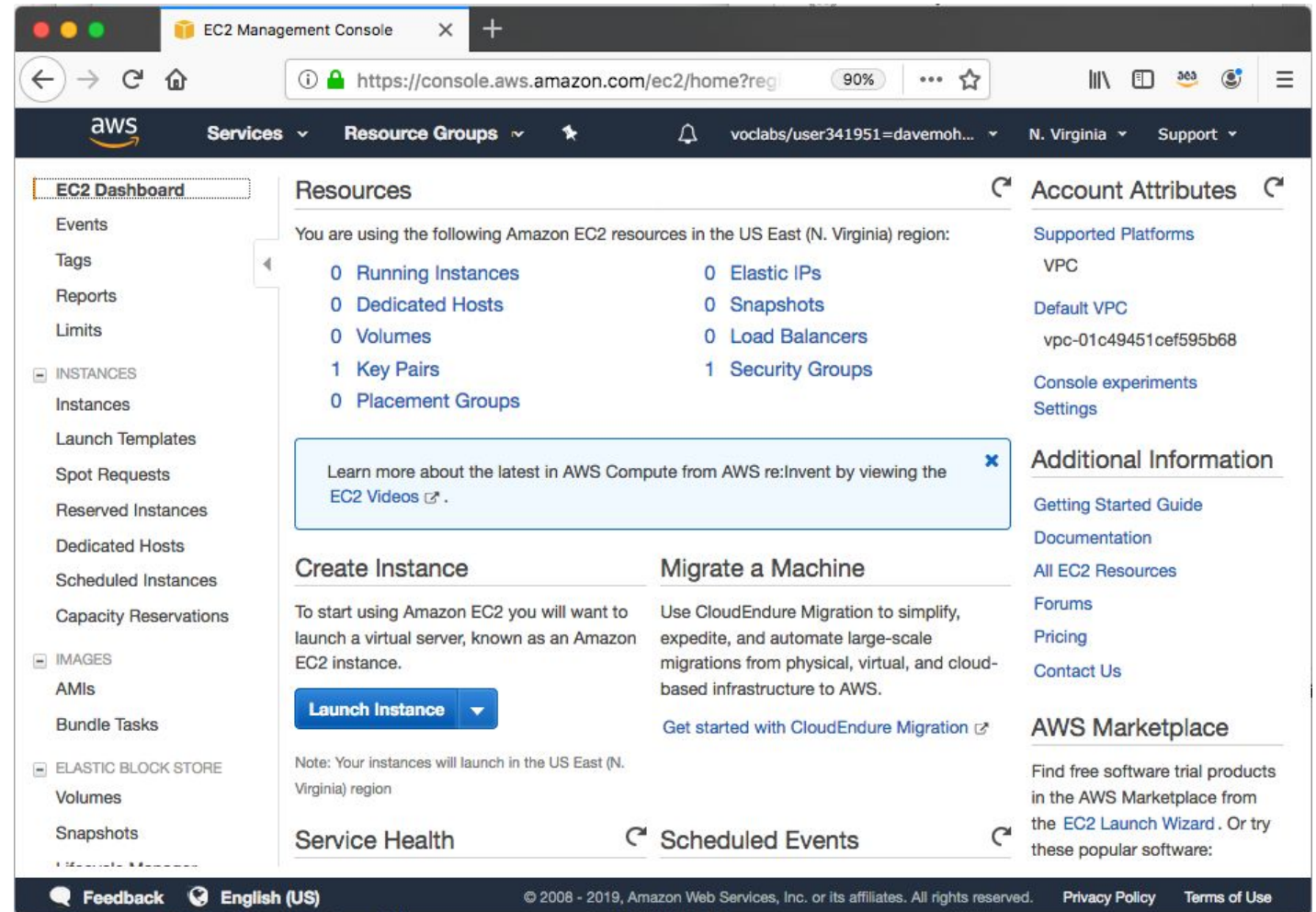


- **Amazon Elastic Compute Cloud (Amazon EC2)**
 - Provides **virtual machines**—referred to as **EC2 instances**—in the cloud.
 - Gives you *full control* over the guest operating system (Windows or Linux) on each instance.
- You can launch instances of any size into an Availability Zone anywhere in the world.
 - Launch instances from **Amazon Machine Images (AMIs)**.
 - Launch instances with a few clicks or a line of code, and they are ready in minutes.
- You can control traffic to and from instances.

Launching an Amazon EC2 instance

This section of the module walks through **nine key decisions** to make when you create an EC2 instance by using the AWS Management Console **Launch Instance Wizard**.

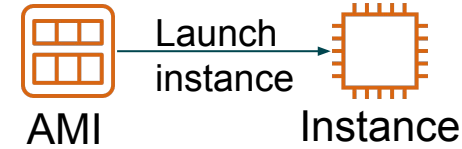
□ Along the way, essential Amazon EC2 concepts will be explored.



1. Select an AMI

Choices made using the Launch Instance Wizard:

1. **AMI**
2. **Instance Type**
3. **Network settings**
4. **IAM role**
5. **User data**
6. **Storage options**
7. **Tags**
8. **Security group**
9. **Key pair**

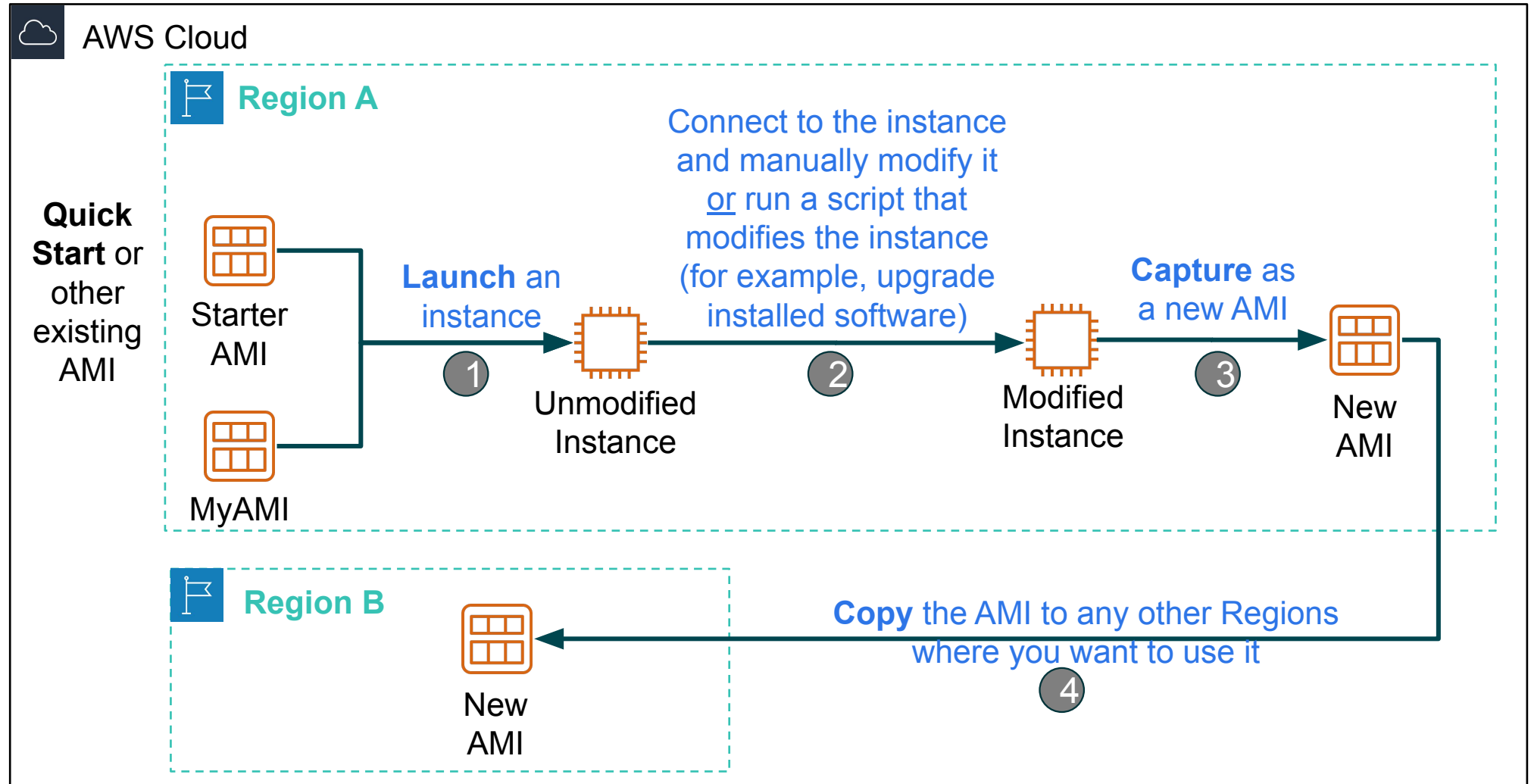


- Amazon Machine Image (AMI)
 - Is a template that is used to create an EC2 instance (which is a **virtual machine, or VM**, that runs in the AWS Cloud)
 - Contains a **Windows** or **Linux** operating system
 - Often also has some **software** pre-installed
- AMI choices:
 - Quick Start – *Linux and Windows AMIs that are provided by AWS*
 - My AMIs – *Any AMIs that you created*
 - AWS Marketplace – *Pre-configured templates from third parties*
 - Community AMIs – *AMIs shared by others; use at your own risk*

Creating a new AMI: Example

AMI details

(Optional) Import a virtual machine

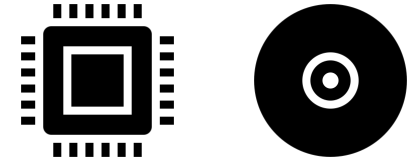


2. Select an instance type

Choices made using the Launch Instance Wizard:

1. AMI
2. **Instance Type**
3. Network settings
4. IAM role
5. User data
6. Storage options
7. Tags
8. Security group
9. Key pair

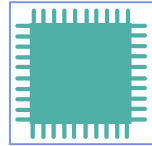
- Consider your use case
 - How will the EC2 instance you create be used?
- The **instance type** that you choose determines –
 - Memory (RAM)
 - Processing power (CPU)
 - Disk space and disk type (Storage)
 - Network performance
- Instance type categories –
 - General purpose
 - Compute optimized
 - Memory optimized
 - Storage optimized
 - Accelerated computing
- Instance types offer *family*, *generation*, and *size*



Select instance type: Based on use case



**General
Purpose**



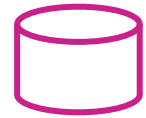
**Compute
Optimized**



**Memory
Optimized**



**Accelerated
Computing**



**Storage
Optimized**

Instance Types	a1, m4, m5, t2, t3	c4, c5	r4, r5, x1, z1	f1, g3, g4, p2, p3	d2, h1, i3
Use Case	Broad	High performance	In-memory databases	Machine learning	Distributed file systems

Instance types: Networking features

- The network bandwidth (Gbps) varies by instance type.
 - See [Amazon EC2 Instance Types](#) to compare.
- To maximize networking and bandwidth performance of your instance type:
 - If you have interdependent instances, launch them into a **cluster placement group**.
 - Enable enhanced networking.
- Enhanced networking types are supported on most instance types.
 - See the [Networking and Storage Features](#) documentation for details.
- Enhanced networking types –
 - **Elastic Network Adapter (ENA)**: Supports network speeds of up to 100 Gbps.
 - **Intel 82599 Virtual Function interface**: Supports network speeds of up to 10 Gbps.

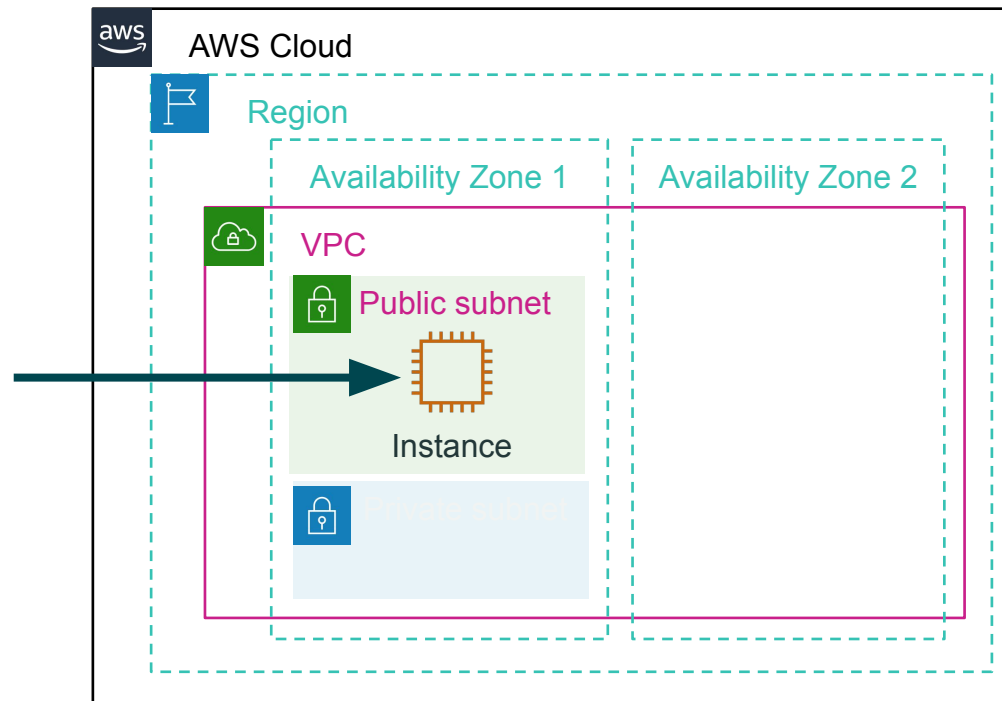
3. Specify network settings

Choices made by using the Launch Instance Wizard:

1. AMI
2. Instance Type
3. Network settings
4. IAM role
5. User data
6. Storage options
7. Tags
8. Security group
9. Key pair

- Where should the instance be deployed?
 - Identify the **VPC** and optionally the **subnet**
- Should a **public IP address** be automatically assigned?
 - To make it internet-accessible

*Example:
specify to
deploy the
instance here*



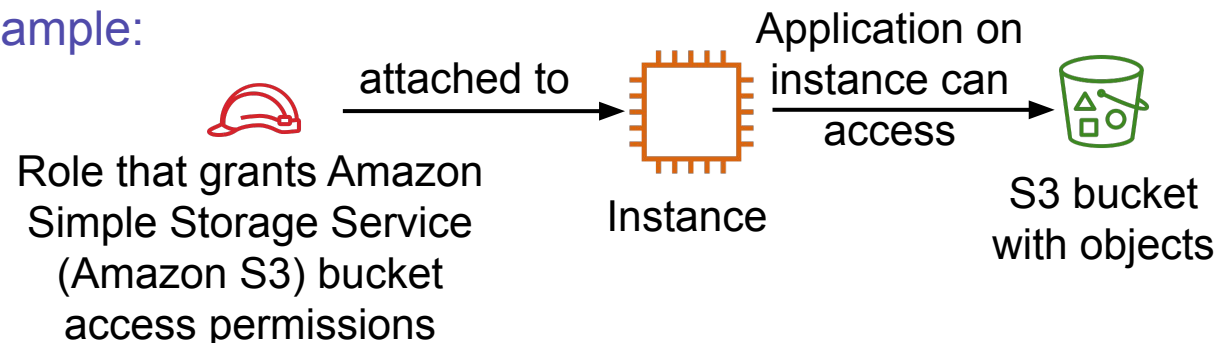
4. Attach IAM role (optional)

Choices made by using the Launch Instance Wizard:

1. AMI
2. Instance Type
3. Network settings
4. **IAM role**
5. User data
6. Storage options
7. Tags
8. Security group
9. Key pair

- Will software on the EC2 instance need to interact with other AWS services?
 - If yes, attach an appropriate **IAM Role**.
- An AWS Identity and Access Management (IAM) role that is attached to an EC2 instance is kept in an **instance profile**.
- You are *not* restricted to attaching a role only at instance launch.
 - You can also attach a role to an instance that already exists.

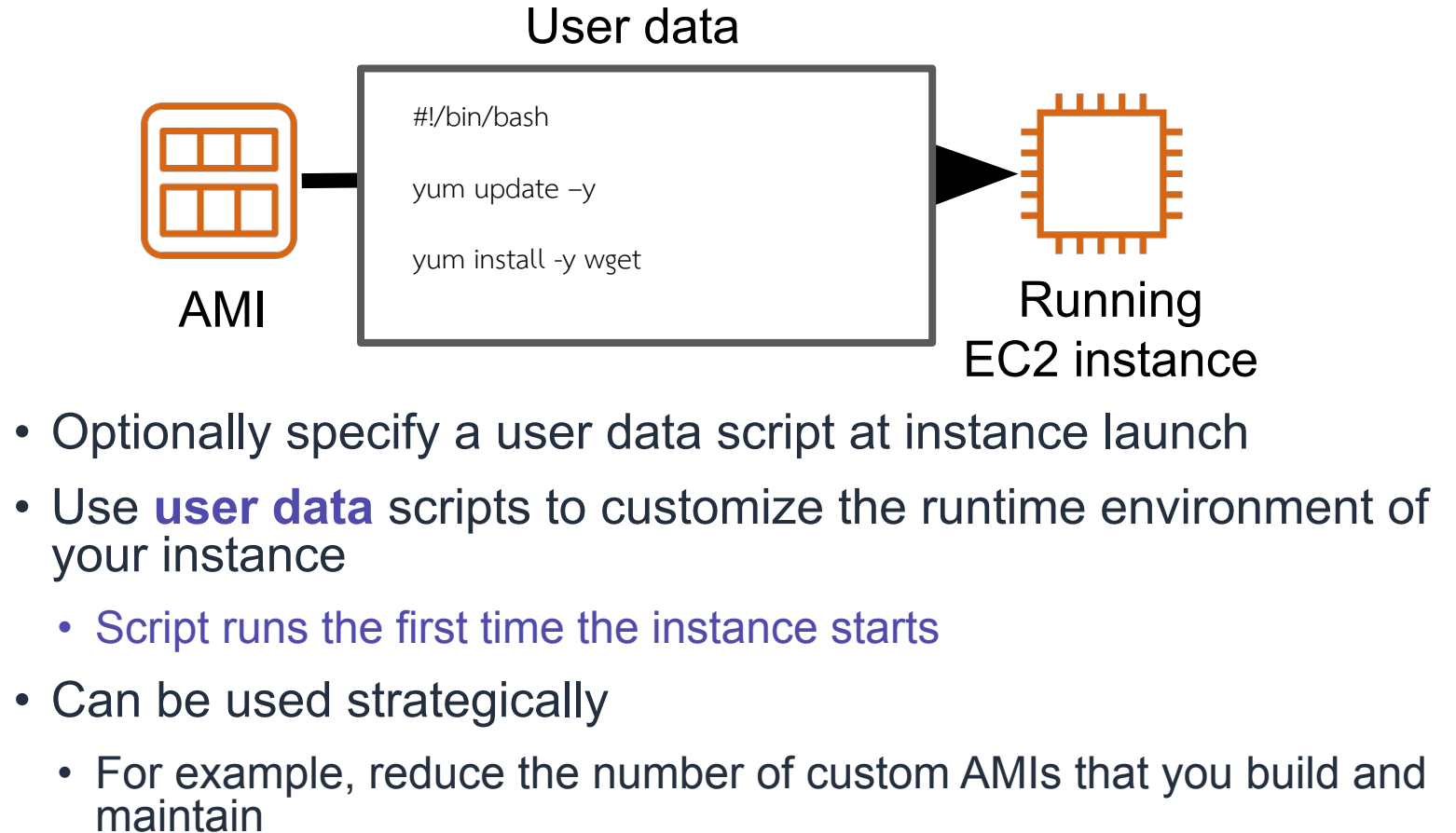
Example:



5. User data script (optional)

Choices made by using the Launch Instance Wizard:

1. AMI
2. Instance Type
3. Network settings
4. IAM role
5. **User data**
6. Storage options
7. Tags
8. Security group
9. Key pair



6. Specify storage

Choices made by using the Launch Instance Wizard:

1. AMI
2. Instance Type
3. Network settings
4. IAM role
5. User data
6. Storage options
7. Tags
8. Security group
9. Key pair

- Configure the **root volume**
 - Where the guest operating system is installed
- Attach **additional storage volumes** (optional)
 - AMI might already include more than one volume
- For each volume, specify:
 - The **size** of the disk (in GB)
 - The **volume type**
 - Different types of solid state drives (SSDs) and hard disk drives (HDDs) are available
 - If the volume will be deleted when the instance is terminated
 - If **encryption** should be used



Amazon EC2 storage options

- **Amazon Elastic Block Store (Amazon EBS)** –

- Durable, block-level storage volumes.
- You can stop the EC2 instance and start it again, and the data will still be there.

- **Amazon EC2 Instance Store** –

- Ephemeral (temporary) storage is provided on disks that are attached to the host computer where the EC2 instance is running.
- If the instance stops, data stored here is deleted.

- Other options for storage (not for the root volume) –

- Mount an **Amazon Elastic File System (Amazon EFS)** file system.
- Connect to **Amazon Simple Storage Service (Amazon S3)**.

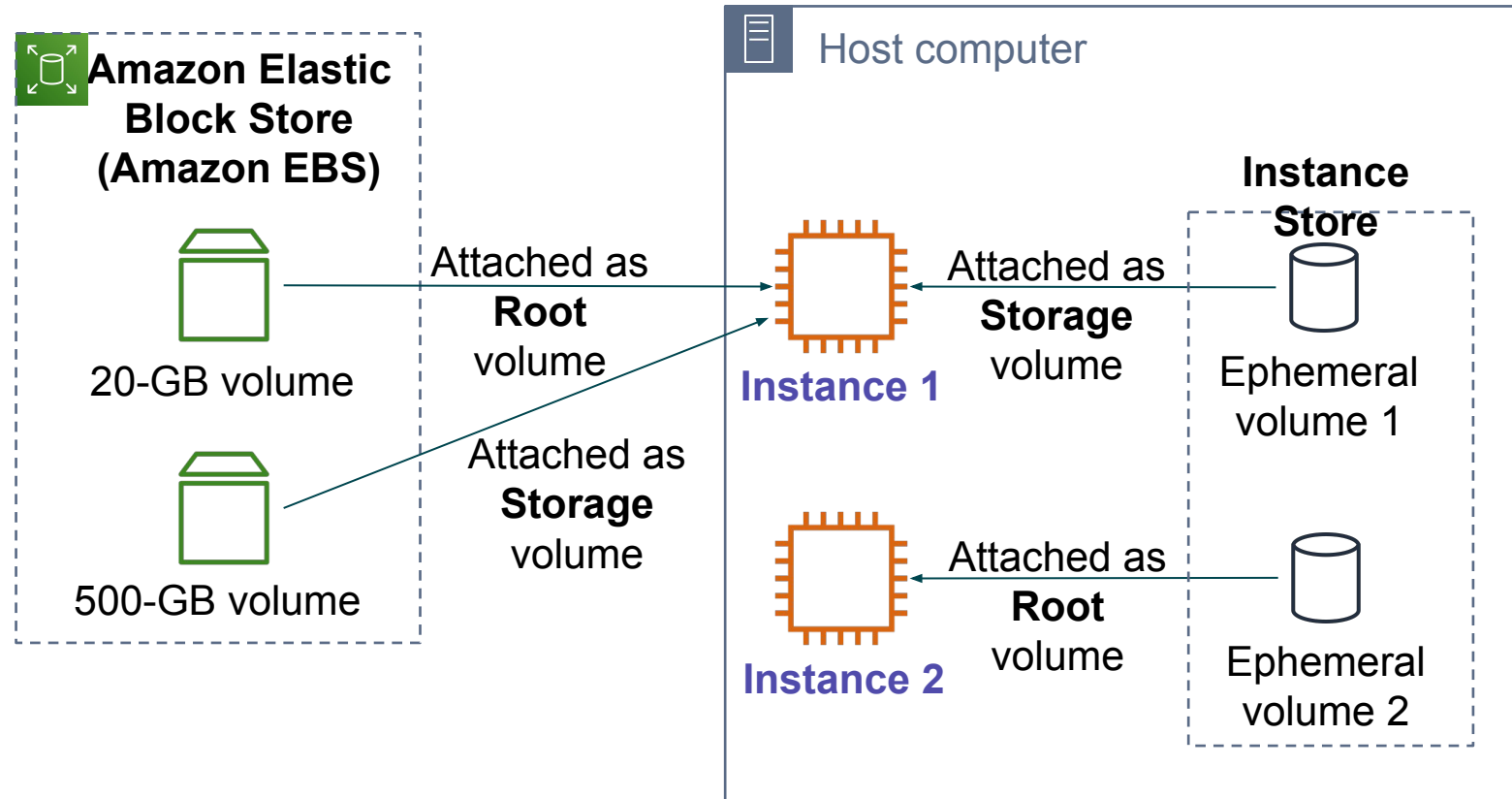
Example storage options

- **Instance 1** characteristics –

- It has an **Amazon EBS** *root volume* type for the operating system.
- What will happen if the instance is stopped and then started again?

- **Instance 2** characteristics –

- It has an **Instance Store** *root volume* type for the operating system.
- What will happen if the instance stops (because of user error or a system malfunction)?



7. Add tags

Choices made by using the Launch Instance Wizard:

1. AMI
2. Instance Type
3. Network settings
4. IAM role
5. User data
6. Storage options
7. Tags
8. Security group
9. Key pair

- A **tag** is a label that you can assign to an AWS resource.
 - Consists of a *key* and an optional *value*.
- Tagging is how you can attach **metadata** to an EC2 instance.
- Potential benefits of tagging—Filtering, automation, cost allocation, and access control.

Example:

Key (128 characters maximum)	Value (256 characters maximum)
Name	WebServer1
Add another tag (Up to 50 tags maximum)	

8. Security group settings

Choices made by using the Launch Instance Wizard:

1. AMI
2. Instance Type
3. Network settings
4. IAM role
5. User data
6. Storage options
7. Tags
8. **Security group**
9. Key pair

- A **security group** is a **set of firewall rules** that control traffic to the instance.
 - It exists *outside* of the instance's guest OS.
- Create **rules** that specify the **source** and which **ports** that network communications can use.
 - Specify the **port** number and the **protocol**, such as Transmission Control Protocol (TCP), User Datagram Protocol (UDP), or Internet Control Message Protocol (ICMP).
 - Specify the **source** (for example, an IP address or another security group) that is allowed to use the rule.

Example rule:

Type ⓘ	Protocol ⓘ	Port Range ⓘ	Source ⓘ
SSH ⌵	TCP	22	My IP ⌵ 72.21.198.67/32

9. Identify or create the key pair

Choices made by using the Launch Instance Wizard:

1. AMI
2. Instance Type
3. Network settings
4. IAM role
5. User data
6. Storage options
7. Tags
8. Security group
9. Key pair

- At instance launch, you specify an existing key pair *or* create a new key pair.
- A **key pair** consists of –
 - A **public key** that AWS stores.
 - A **private key** file that you store.
- It enables secure connections to the instance.
- For **Windows AMIs** –
 - Use the private key to obtain the administrator password that you need to log in to your instance.
- For **Linux AMIs** –
 - Use the private key to use SSH to securely connect to your instance.



mykey.pem



Amazon EC2 console view of a running EC2 instance

The screenshot displays the Amazon EC2 console interface. The left sidebar contains navigation links for EC2 Dashboard, Events, Tags, Reports, Limits, INSTANCES, Launch Templates, Spot Requests, Reserved Instances, Dedicated Hosts, Scheduled Instances, Capacity Reservations, IMAGES, AMIs, Bundle Tasks, ELASTIC BLOCK STORE, Volumes, and Snapshots. The main content area shows a list of instances with a search bar and filters. A table lists the instance details, including Name, Instance ID, Instance Type, Instance State, Status Checks, Public DNS (IPv4), and IPv4 Public IP. The instance i-092b6f3efba959a53 is shown in a running state. Below the table, the instance details are expanded, showing the Description, Status Checks, Monitoring, and Tags tabs. The Description tab is active, displaying various attributes such as Instance ID, Instance state, Instance type, Elastic IPs, Availability zone, Security groups, Scheduled events, AMI ID, Platform, Public DNS (IPv4), IPv4 Public IP, IPv6 IPs, Private DNS, Private IPs, Secondary private IPs, VPC ID, Subnet ID, and Network interfaces.

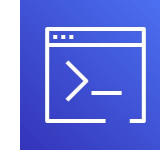
Name	Instance ID	Instance Type	Instance State	Status Checks	Public DNS (IPv4)	IPv4 Public IP
	i-092b6f3efba959a53	t2.micro	running	Initializing	ec2-54-159-171-63.co...	54.159.171.63

Instance: i-092b6f3efba959a53 Public DNS: ec2-54-159-171-63.compute-1.amazonaws.com

Description		Status Checks	Monitoring	Tags
Instance ID	i-092b6f3efba959a53			
Instance state	running			
Instance type	t2.micro			
Elastic IPs				
Availability zone	us-east-1c			
Security groups	launch-wizard-1. view inbound rules view outbound rules			
Scheduled events	No scheduled events			
AMI ID	amzn2-ami-hvm-2.0.20190823.1-x86_64-gp2 (ami-0b69ea66ff7391e80)			
Platform	-			
Public DNS (IPv4)	ec2-54-159-171-63.compute-1.amazonaws.com			
IPv4 Public IP	54.159.171.63			
IPv6 IPs	-			
Private DNS	ip-172-31-82-44.ec2.internal			
Private IPs	172.31.82.44			
Secondary private IPs				
VPC ID	vpc-e4e9859e			
Subnet ID	subnet-d22779fc			
Network interfaces	eth0			

Another option: Launch an EC2 instance with the AWS Command Line Interface

- EC2 instances can also be created programmatically.



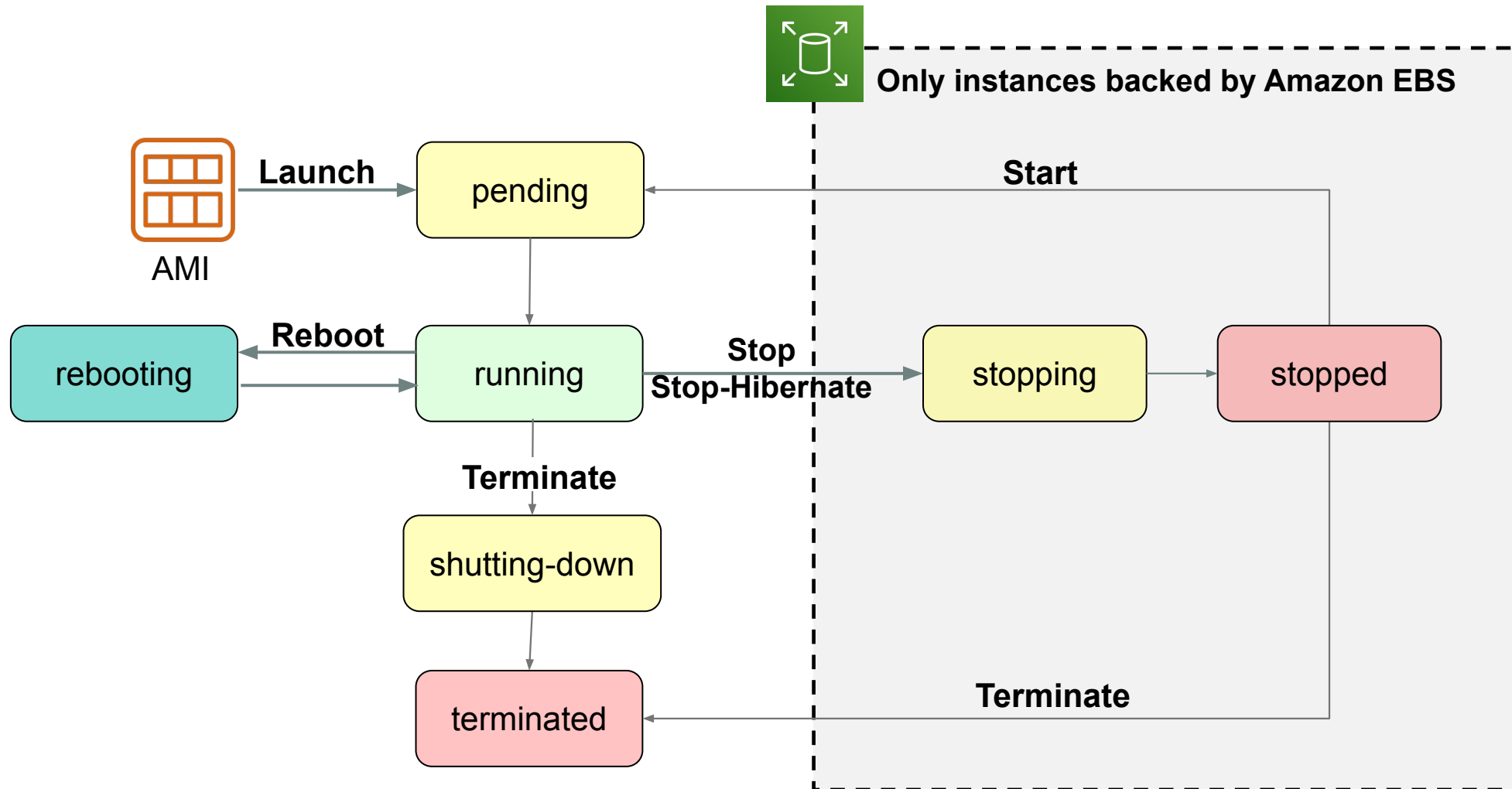
AWS Command Line Interface (AWS CLI)

- This example shows how simple the command can be.
 - This command assumes that the key pair and security group already exist.
- More options could be specified. See the [AWS CLI Command Reference](#) for details.

Example command:

```
aws ec2 run-instances \  
--image-id ami-1a2b3c4d \  
--count 1 \  
--instance-type c3.large \  
--key-name MyKeyPair \  
--security-groups MySecurityGroup \  
--region us-east-1
```

Amazon EC2 instance lifecycle



Consider using an Elastic IP address

- **Rebooting** an instance will *not* change any IP addresses or DNS hostnames.
- When an instance is **stopped** and then **started** again –
 - The *public* IPv4 address and *external* DNS hostname will change.
 - The *private* IPv4 address and internal DNS hostname do *not* change.
- If you require a persistent public IP address –
 - Associate an **Elastic IP address** with the instance.
- Elastic IP address characteristics –
 - Can be associated with instances in the Region as needed.
 - Remains allocated to your account until you choose to release it.



Elastic IP
Address

EC2 instance metadata

- **Instance metadata** is data about your instance.
- While you are connected to the instance, you can view it –
 - In a browser: `http://169.254.169.254/latest/meta-data/`
 - In a terminal window: `curl http://169.254.169.254/latest/meta-data/`
- Example retrievable values –
 - Public IP address, private IP address, public hostname, instance ID, security groups, Region, Availability Zone.
 - Any user data specified at instance launch can also be accessed at:
`http://169.254.169.254/latest/user-data/`
- It can be used to configure or manage a running instance.
 - For example, author a configuration script that reads the metadata and uses it to configure applications or OS settings.

Amazon CloudWatch for monitoring

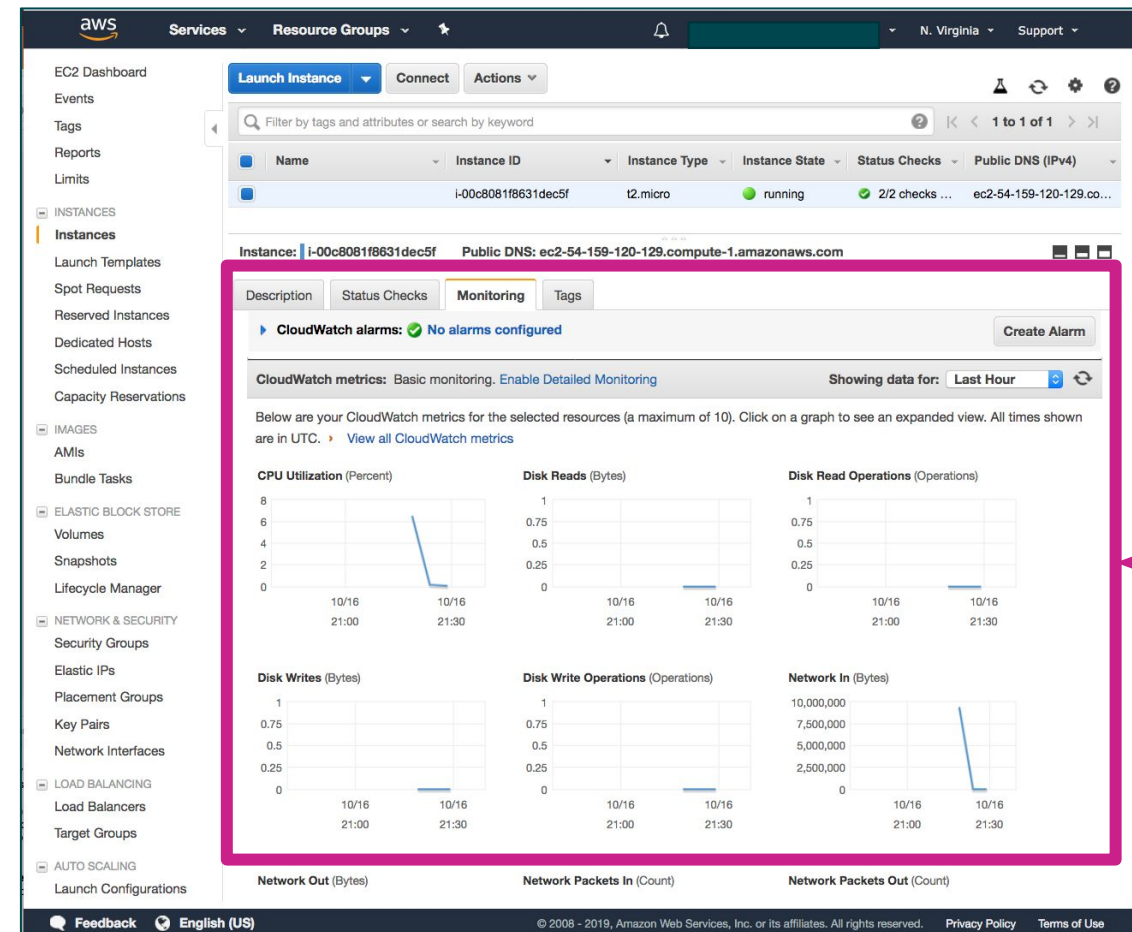
- Use **Amazon CloudWatch** to monitor EC2 instances
 - Provides near-real-time metrics
 - Provides charts in the Amazon EC2 console **Monitoring** tab that you can view
 - Maintains 15 months of historical data
- **Basic monitoring**
 - Default, no additional cost
 - Metric data sent to CloudWatch every 5 minutes
- **Detailed monitoring**
 - Fixed monthly rate for seven pre-selected metrics
 - Metric data delivered every 1 minute



Amazon CloudWatch



Instance with CloudWatch



Section 2 key takeaways



- **Amazon EC2** enables you to run Windows and Linux **virtual machines** in the cloud.
- You launch **EC2 instances** from an **AMI** template into a VPC in your account.
- You can choose from many **instance types**. Each instance type offers different combinations of CPU, RAM, storage, and networking capabilities.
- You can configure **security groups** to control access to instances (specify allowed ports and source).
- **User data** enables you to specify a script to run the first time that an instance launches.
- Only **instances that are backed by Amazon EBS** can be stopped.
- You can use **Amazon CloudWatch** to capture and review metrics on EC2 instances.

Recorded Amazon EC2 demonstration



Set up demo

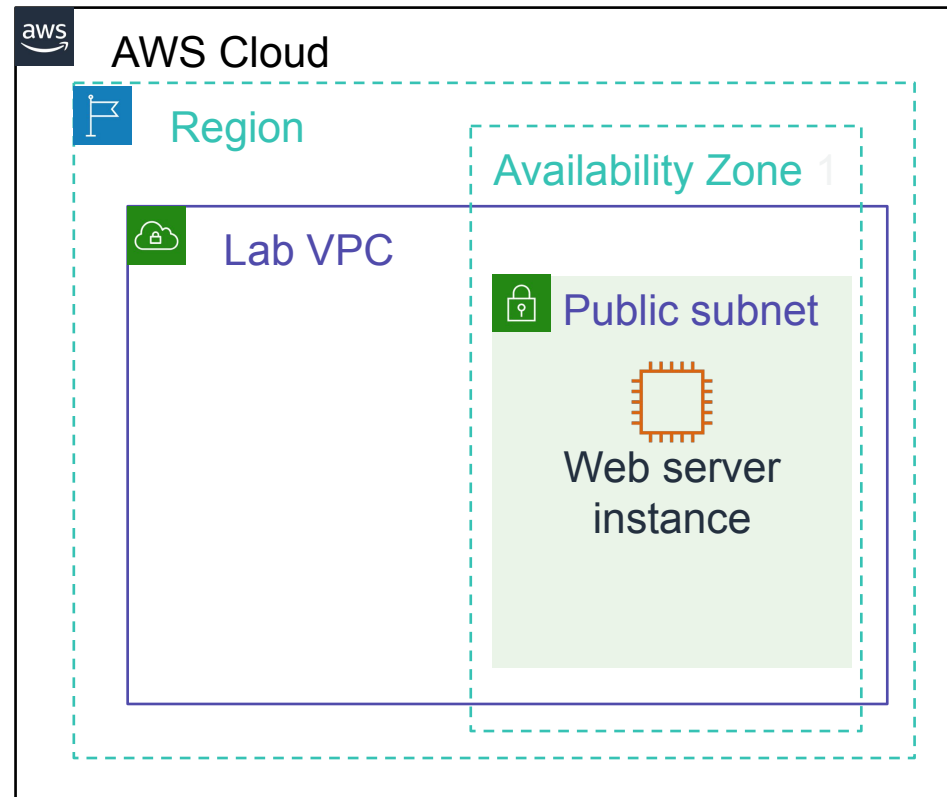
Amazon Elastic Compute Cloud
(Amazon EC2)

Lab 3: Introduction to Amazon EC2



Lab 3 scenario

In this lab, you will launch and configure your first virtual machine that runs on Amazon EC2.



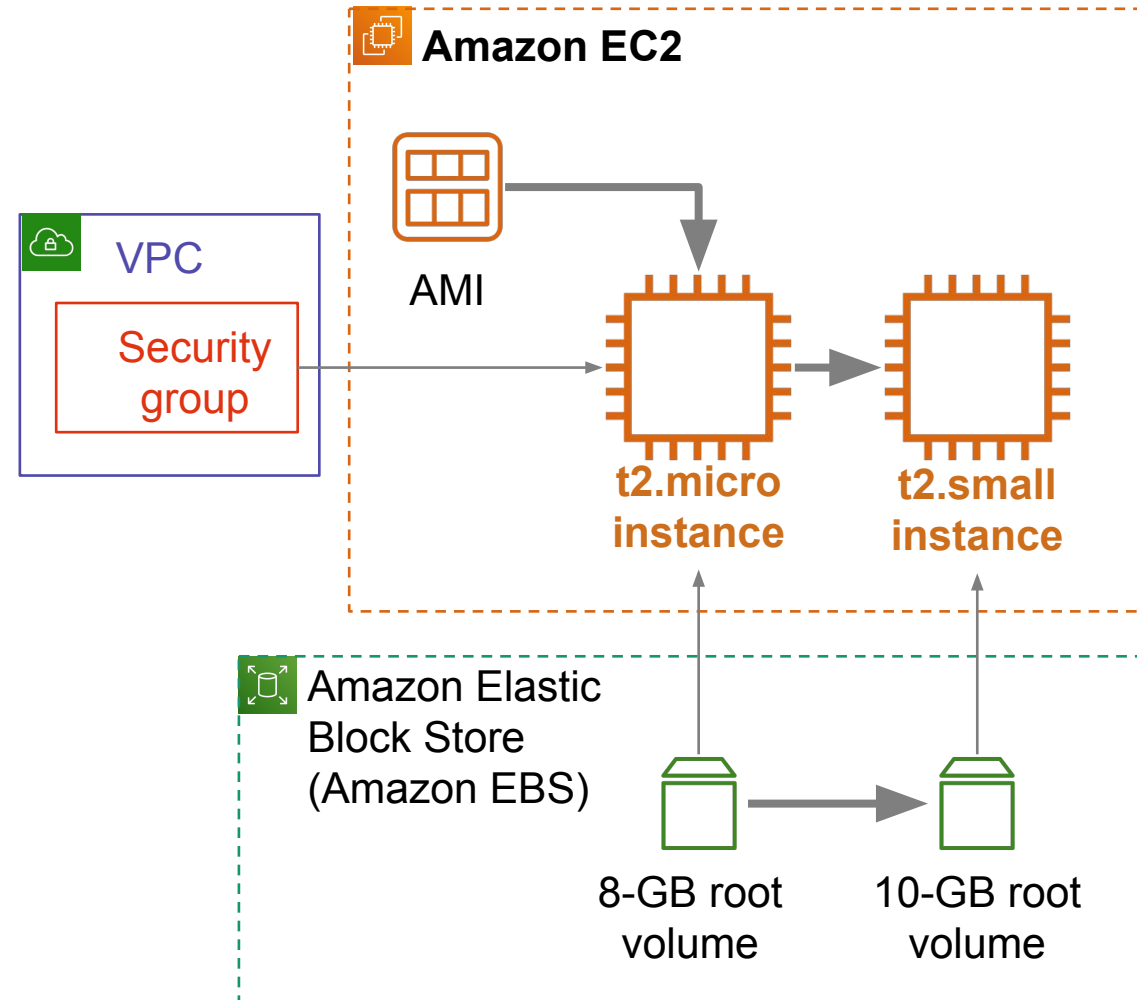
Lab 3: Tasks

- Task 1 – Launch Your Amazon EC2 Instance
- Task 2 – Monitor Your Instance
- Task 3 – Update Your Security Group and Access the Web Server
- Task 4 – Resize Your Instance: Instance Type and EBS Volume
- Task 5 – Explore EC2 Limits
- Task 6 – Test Termination Protection

Lab 3: Final product

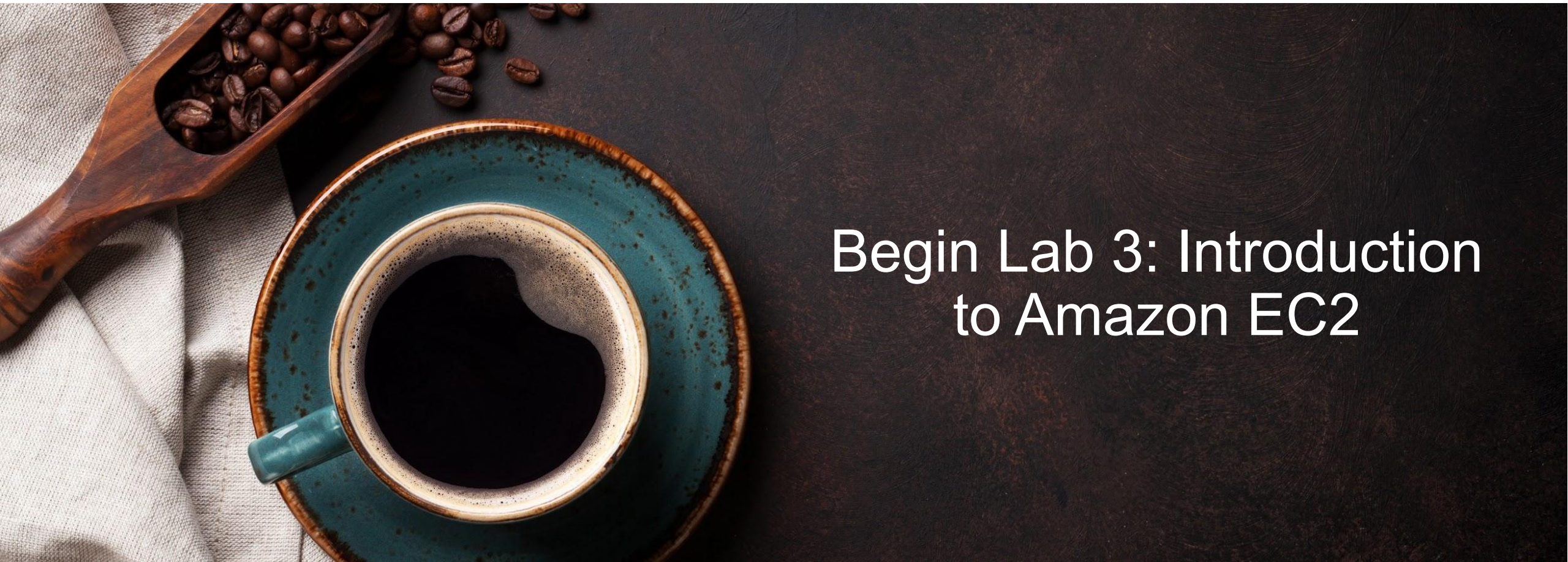
By the end of the lab, you will have:

1. Launched an instance that is configured as a web server
2. Viewed the instance system log
3. Reconfigured a security group
4. Modified the instance type and root volume size





~ 35 minutes



Begin Lab 3: Introduction to Amazon EC2

Lab debrief: Key takeaways



Activity 2: "Lab 3 Understanding Check" (10-15 minutes)

Instructions

Q1: Explain the purpose of using a User Data script when launching an EC2 instance and provide an example from the lab.

Q2: Why was the web server initially inaccessible after launching the EC2 instance, and how was this issue resolved?

Q3: What steps are involved in resizing an Amazon EC2 instance, and what additional action must be taken before resizing?

Q4: How can you view the system log and the console output of an EC2 instance, and what is the importance of these tools?

Q5: What happens if you attempt to stop an EC2 instance with stop protection enabled, and how can you successfully stop the instance?

Q6: What are the default storage settings for the EC2 instance in this lab, and why might you need to modify the storage size?

Activity 3:

"EC2 Deploy & Dominate"

(15-20 minutes)

Instructions:

- Each group (3-4 ppl) will create an EC2 instance that meets the following requirements:
 - Cheapest general-purpose instance with Intel CPU (2 vCPUs, and 1GB RAM)
On-Demand hourly pricing
 - Linux OS that has AWS CLI installed (test with `aws --version` at the shell)
 - Allows SSH access for remote maintenance
 - Configured as a basic web server
- Each group will prepare to present:
 - 1 How they selected the correct EC2 instance type
 - 2 How they secured SSH access
 - 3 What challenges they faced and how they solved them

END